

Optimization Theory: A Briefing

1 Optimization: examples

1. The general consumer optimization problem:

$$\begin{aligned} & \max u(\mathbf{x}) \\ & s.t. \quad \mathbf{p}\mathbf{x} \leq I \\ & \quad \mathbf{x} \geq \mathbf{0} \end{aligned}$$

where $\mathbf{x} \in \mathbb{R}_+^n$ represents a consumption bundle, and I the income of the consumer.

2. Or the dual

$$\begin{aligned} & \min \mathbf{p}\mathbf{x} \\ & s.t. \quad u(\mathbf{x}) \geq U_0, \\ & \quad \mathbf{x} \geq \mathbf{0} \end{aligned}$$

3. The firm's problem:

$$\max \pi(\mathbf{x}) = \mathbf{p}f(\mathbf{x}) - c(\mathbf{x}), \quad \mathbf{x} \geq \mathbf{0}$$

4. Or

$$\begin{aligned} & \min c(\mathbf{x}) \\ & s.t. \quad f(\mathbf{x}) \geq y, \\ & \quad \mathbf{x} \geq \mathbf{0} \end{aligned}$$

2 Unconstrained Optimization

2.1 Single variable

1. Consider the function, $y = f(x)$. The function achieves a local maximum at x^* if $f(x^*) \geq f(x)$ for all x in some neighborhood of x^* ; the function achieves a global maximum at x^* if $f(x^*) \geq f(x)$ for all x in the domain of the function.
2. If $f'(x^*) = 0$ and $f''(x^*) < 0$, then $f(x)$ reach a local maximum at x^* . If $f'(\tilde{x}) = 0$ and $f''(\tilde{x}) > 0$, then $f(x)$ reach a local minimum at $\tilde{x}$.

2.2 Several variables

1. (J&R, Theorem A2.9). FONC (first-order necessary condition). If the differentiable function $f(\mathbf{x})$ reaches a local interior maximum and minimum at $\mathbf{x}^*$, then $\mathbf{x}^*$ solves the systems of simultaneous equations,

$$\frac{\partial f(\mathbf{x}^*)}{\partial x_1} = 0, \frac{\partial f(\mathbf{x}^*)}{\partial x_2} = 0, \dots, \frac{\partial f(\mathbf{x}^*)}{\partial x_n} = 0.$$

2. (J&R, Theorem A2.10). SONC (second-order necessary condition). Let $f(\mathbf{x})$ be twice continuously differentiable. If $f(\mathbf{x})$ reaches a local interior maximum (minimum) at $\mathbf{x}^*$, then $H(\mathbf{x}^*)$ is negative (positive) semidefinite.
3. (J&R, Theorem A2.12). Sufficient conditions. If $f_i(\mathbf{x}^*) = 0$ for $i = 1, \dots, n$ and $H(\mathbf{x}^*)$ is negative (positive) definite at $\mathbf{x}^*$, then $f(\mathbf{x})$ reaches a local maximum (minimum) at $\mathbf{x}^*$.
4. (J&R, Theorem A2.13). Local-Global Theorem. Let f be a twice continuously differentiable real-valued **concave** function on D . The following statement are equivalent, where $\mathbf{x}^*$ is an interior point of D : $\nabla f(\mathbf{x}^*) = 0$; f achieves a local maximum at $\mathbf{x}^*$; f achieves a global maximum at $\mathbf{x}^*$. (First order condition is sufficient now).
5. (J&R, Theorem A2.14). If $\mathbf{x}^*$ maximizes (minimizes) the **strictly concave** (convex) function f , then $\mathbf{x}^*$ is the **unique** global maximizer (minimizer).
6. (J&R, Theorem A2.15). Sufficient conditions. Let $f(\mathbf{x})$ be twice continuously differentiable. If $f(\mathbf{x})$ is **strictly concave** (**convex**) and $f_i(\mathbf{x}^*) = 0$, $i = 1, \dots, n$, then $\mathbf{x}^*$ is the unique global maximizer (minimizer) of $f(\mathbf{x})$.

2.3 Example

1. A monopolist producing a single output has two types of customers. Type 1 customers are willing to pay a price $p_1 = 50 - 5Q_1$, type 2s willing to pay $p_2 = 100 - 10Q_2$. The cost is $90 + 20Q$. How much should the monopolist produce for each market?
2. The monopolist's problem is

$$\max_{\{Q_1, Q_2\}} \pi = Q_1(50 - 5Q_1) + Q_2(100 - 10Q_2) - (90 + 20(Q_1 + Q_2))$$

3. First-order conditions:

$$\begin{aligned}\frac{\partial \pi}{\partial Q_1} &= 50 - 10Q_1 - 20 = 0, \text{ or } Q_1 = 3, \\ \frac{\partial \pi}{\partial Q_2} &= 100 - 20Q_2 - 20 = 0, \text{ or } Q_2 = 4.\end{aligned}$$

4. Now check the second order conditions:

$$\pi_{Q_1 Q_1} = -10, \pi_{Q_2 Q_2} = -20, \pi_{Q_1 Q_2} = \pi_{Q_2 Q_1} = 0.$$

5. The Hessian Matrix is negative definite, therefore π is strictly concave. Hence, $(3, 4)$ is unique global profit-maximizing supply plan.

3 Constrained Optimization

3.1 Equality constraints

3.1.1 Two variables and one equality constraint

1. The problem is

$$\max_{x_1, x_2} f(x_1, x_2) \quad \text{subject to } g(x_1, x_2) = c.$$

2. One way to solve it is by substitution. For example, if $g(x_1, x_2) = c$ can be rewritten as $x_2 = \tilde{g}(x_1)$. We can substitute it to the original problem, then we have no more constraint now:

$$\max_{x_1} f(x_1, \tilde{g}(x_1)).$$

3. Lagrange's method.

$$L(x_1, x_2, \lambda) = f(x_1, x_2) + \lambda(c - g(x_1, x_2))$$

4. First-order conditions are

$$\begin{aligned}\frac{\partial L}{\partial x_1} &= \frac{\partial f}{\partial x_1} - \lambda \frac{\partial g}{\partial x_1} = 0 \\ \frac{\partial L}{\partial x_2} &= \frac{\partial f}{\partial x_2} - \lambda \frac{\partial g}{\partial x_2} = 0 \\ \frac{\partial L}{\partial \lambda} &= c - g(x_1, x_2) = 0\end{aligned}$$

5. (J&R, Theorem A2.17) Sufficient condition. If $(x_1^*, x_2^*, \lambda^*)$ solves the first-order conditions, and if $|\overline{H}| > 0 (< 0)$ when evaluated at $(x_1^*, x_2^*, \lambda^*)$, then (x_1^*, x_2^*) is a local maximum (minimum), where

$$\overline{H} = \begin{pmatrix} 0 & g_1 & g_2 \\ g_1 & L_{11} & L_{12} \\ g_2 & L_{21} & L_{22} \end{pmatrix}$$

6. Example. Consider the following problem:

$$\begin{aligned}\max \quad & f(x, y) = x^2 y \\ \text{s.t.} \quad & 2x^2 + y^2 = 3\end{aligned}$$

7. Form the Lagrangian

$$L(x, y, \lambda) = x^2 y + \lambda(3 - 2x^2 - y^2).$$

The first-order conditions are:

$$\begin{aligned}\frac{\partial L}{\partial x} &= 2xy - 4\lambda x = 2x(y - 2\lambda) = 0 \\ \frac{\partial L}{\partial y} &= x^2 - 2\lambda y = 0 \\ \frac{\partial L}{\partial \lambda} &= 3 - 2x^2 - y^2 = 0.\end{aligned}$$

Solving the above equations, we have six solutions $(0, \sqrt{3}, 0)$, $(0, -\sqrt{3}, 0)$; $(1, 1, 0.5)$, $(1, -1, -0.5)$, $(-1, -1, -0.5)$, $(-1, 1, 0.5)$.

8. Since

$$\begin{aligned}f(1, 1) &= f(-1, 1) = 1 \\ f(1, -1) &= f(-1, -1) = -1 \\ f(0, \sqrt{3}) &= f(0, -\sqrt{3}) = 0,\end{aligned}$$

the max occurs at $(-1, 1)$ and $(1, 1)$. And the min occurs at $(1, -1)$ and $(-1, -1)$.

3.1.2 Several Equality Constraints

1. We consider the problem of maximizing a function $f(x_1, \dots, x_n)$ of n variables constrained by m equality constraints. Let the functions $g_1(\mathbf{x}), \dots, g_m(\mathbf{x})$ define the constraint set. We want to

$$\begin{aligned} & \max \text{ or } \min f(x_1, \dots, x_n) \\ & s.t. \ g_1(\mathbf{x}) = a_1, \dots, g_m(\mathbf{x}) = a_m. \end{aligned}$$

2. Lagrange's method.

$$L(\mathbf{x}, \boldsymbol{\lambda}) = f(\mathbf{x}) + \lambda_1(a_1 - g_1(\mathbf{x})) + \lambda_2(a_2 - g_2(\mathbf{x})) + \dots + \lambda_m(a_m - g_m(\mathbf{x}))$$

3. First-order conditions are

$$\begin{aligned} \frac{\partial L}{\partial x_1}(\mathbf{x}^*, \boldsymbol{\lambda}^*) &= 0, \dots, \frac{\partial L}{\partial x_n}(\mathbf{x}^*, \boldsymbol{\lambda}^*) = 0 \\ \frac{\partial L}{\partial \lambda_1}(\mathbf{x}^*, \boldsymbol{\lambda}^*) &= 0, \dots, \frac{\partial L}{\partial \lambda_m}(\mathbf{x}^*, \boldsymbol{\lambda}^*) = 0 \end{aligned}$$

4. Example. Consider the following problem:

$$\begin{aligned} & \max f(x, y, z) = xyz \\ & s.t \ x^2 + y^2 = 1 \\ & \quad x + z = 1 \end{aligned}$$

5. First, form the Lagrangian:

$$L(x, y, z, \lambda_1, \lambda_2) = xyz + \lambda_1(1 - x^2 - y^2) + \lambda_2(1 - x - z).$$

6. Second, the first-order conditions are:

$$\begin{aligned} \frac{\partial L}{\partial x} &= yz - 2\lambda_1 x - \lambda_2 = 0 \\ \frac{\partial L}{\partial y} &= xz - 2\lambda_1 y = 0 \\ \frac{\partial L}{\partial z} &= xy - \lambda_2 = 0 \\ \frac{\partial L}{\partial \lambda_1} &= 1 - x^2 - y^2 = 0 \\ \frac{\partial L}{\partial \lambda_2} &= 1 - x - z = 0. \end{aligned}$$

7. Solving the above equation system, we obtain

$$\begin{aligned}x &\simeq 0.4343, \ y = \pm 0.9008, \ z = 0.5657; \\x &\simeq -0.7676, \ y = \pm 0.6409, \ z = 1.7676.\end{aligned}$$

8. Evaluating the objective function at these four points, we find the maximizer is

$$x \simeq -0.7676, \ y = -0.6409, \ z = 1.7676.$$

3.2 Inequality constraints

3.2.1 One inequality constraint

- The simple case:

$$\max_x f(x) \quad \text{subject to } x \geq 0$$

- Three possible cases:

$$\begin{aligned}x^* = 0 \quad &\text{and } f'(x^*) < 0 \\x^* = 0 \quad &\text{and } f'(x^*) = 0 \\x^* > 0 \quad &\text{and } f'(x^*) = 0\end{aligned}$$

- Summarize the three cases, we have

$$\begin{aligned}f'(x^*) &\leq 0 \\x^* f'(x^*) &= 0 \\x^* &\geq 0\end{aligned}$$

- Two variables and one inequality constraint:

$$\max f(x, y), \quad \text{subject to } h(x, y) \leq b$$

- First, we form the Lagrangian function:

$$L(x, y, \lambda) = f(x, y) + \lambda(b - h(x, y)).$$

- Two possibilities:

1. (a) If the constraint is binding (active, effective, tight), or $h(x, y) = b$, we should have $\lambda \geq 0$

- (b) If the constraint is not binding (inactive, ineffective, loose), or $h(x, y) < b$, we should have $\lambda = 0$.
- (c) In summary of the above two cases, we should have $\lambda[h(x, y) - b] = 0$. This is so-called **complementary slackness condition**.

- Therefore, the first-order conditions are:

$$\begin{aligned}\frac{\partial L}{\partial x}(x^*, y^*, \lambda^*) &= 0 \\ \frac{\partial L}{\partial y}(x^*, y^*, \lambda^*) &= 0 \\ \lambda^*[h(x^*, y^*) - b] &= 0 \\ \lambda^* &\geq 0 \\ h(x^*, y^*) &\leq b\end{aligned}$$

- The similarities and differences between the problems with equality constraints and those with inequality constraints:
 - Both use the Lagrangian and both requires that the derivatives of L w.r.t the x to be 0.
 - The condition that $\frac{\partial L}{\partial \lambda} = 0$ for equality constraint may no longer hold for inequality constraints since the constraint need not be binding at the maximizer in the inequality constraint case. It is replaced by two conditions:

$$\lambda^*[h(x^*, y^*) - b] = 0, \quad h(x^*, y^*) \leq b.$$

- There were no restriction on the sign of the multiplier in the equality constraint case; however, the multiplier for inequality constraints must be nonnegative.
- Example. Consider the following problem:

$$\begin{aligned}\max \quad & f(x, y) = xy \\ \text{s.t.} \quad & x^2 + y^2 \leq 1.\end{aligned}$$

- Form the Lagrangian

$$L(x, y, \lambda) = xy + \lambda(1 - x^2 - y^2).$$

- Then write out the conditions described above:

$$\begin{aligned}\frac{\partial L}{\partial x} &= y - 2\lambda x = 0 \\ \frac{\partial L}{\partial y} &= x - 2\lambda y = 0 \\ \lambda(x^2 + y^2 - 1) &= 0 \\ \lambda &\geq 0 \\ x^2 + y^2 &\leq 1\end{aligned}$$

- The first two equations yield

$$\lambda = \frac{y}{2x} = \frac{x}{2y}, \text{ or } x^2 = y^2 \quad (1)$$

If $\lambda = 0$, then $x = y = 0$. This combination satisfies all the conditions above, so it is a candidate for a solution.

- If $\lambda \neq 0$, then the third equation becomes $x^2 + y^2 - 1 = 0$. Combining with (1), we find the following four candidates:

$$\begin{aligned}x &= +\frac{1}{\sqrt{2}}, y = +\frac{1}{\sqrt{2}}, \lambda = +\frac{1}{2}; \\ x &= -\frac{1}{\sqrt{2}}, y = -\frac{1}{\sqrt{2}}, \lambda = +\frac{1}{2}; \\ x &= +\frac{1}{\sqrt{2}}, y = -\frac{1}{\sqrt{2}}, \lambda = -\frac{1}{2}; \\ x &= -\frac{1}{\sqrt{2}}, y = +\frac{1}{\sqrt{2}}, \lambda = -\frac{1}{2}.\end{aligned}$$

- We disregard the last two candidates since they involve a negative multiplier. So, including (0,0,0), there are three candidates which satisfy all five first order conditions. Plugging these three into the objective function, we find that

$$x = \frac{1}{\sqrt{2}}, y = \frac{1}{\sqrt{2}} \text{ and } x = -\frac{1}{\sqrt{2}}, y = -\frac{1}{\sqrt{2}}$$

are the solutions of our original problem.

3.2.2 Several inequality constraints

1. Suppose there are k inequality constraints, and the problem is

$$\begin{aligned}\max \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & g_1(\mathbf{x}) \leq b_1, \dots, g_k(\mathbf{x}) \leq b_k\end{aligned}$$

2. Form the Lagrangian

$$L(\mathbf{x}, \boldsymbol{\lambda}) = f(\mathbf{x}) + \lambda_1(b_1 - g_1(\mathbf{x})) + \dots + \lambda_k(b_k - g_k(\mathbf{x})).$$

3. Then, there exists multipliers $\lambda_1^*, \dots, \lambda_k^*$ such that

$$\begin{aligned} \frac{\partial L}{\partial x_1}(\mathbf{x}^*, \boldsymbol{\lambda}^*) &= 0, \dots, \frac{\partial L}{\partial x_n}(\mathbf{x}^*, \boldsymbol{\lambda}^*) = 0 \\ \lambda_1^*[g_1(x^*, y^*) - b_1] &= 0, \dots, \lambda_k^*[g_k(x^*, y^*) - b_k] = 0 \\ \lambda_1^* &\geq 0, \dots, \lambda_k^* \geq 0 \\ g_1(\mathbf{x}^*) &\leq b_1, \dots, g_k(\mathbf{x}^*) \leq b_k \end{aligned}$$

4. Example.

$$\begin{aligned} \max \quad & f(x, y, z) = x - y^2 \\ \text{s.t.} \quad & x^2 + y^2 = 4 \\ & x, y \geq 0. \end{aligned}$$

5. First, form the Lagrangian:

$$L = x - y^2 + \lambda_1(4 - x^2 - y^2) + \lambda_2x + \lambda_3y$$

6. Next, we write out the complete set of the first-order conditions:

$$\frac{\partial L}{\partial x} = 1 - 2\lambda_1x + \lambda_2 = 0 \tag{2}$$

$$\frac{\partial L}{\partial y} = -2y - 2\lambda_1y + \lambda_3 = 0 \tag{3}$$

$$x^2 + y^2 - 4 = 0 \tag{4}$$

$$\lambda_2x = 0 \tag{5}$$

$$\lambda_3y = 0 \tag{6}$$

$$\lambda_2 \geq 0$$

$$\lambda_3 \geq 0$$

$$x \geq 0$$

$$y \geq 0$$

7. Write (2) as $1 + \lambda_2 = 2\lambda_1x$. Since $\lambda_2 \geq 0$, $1 + \lambda_2 > 0$. Therefore, $\lambda_1 > 0$ and $x > 0$ (and $\lambda_2 = 0$ by (5)).

8. Write (3) as $2y(1 + \lambda_1) = \lambda_3$. Since $1 + \lambda_1 > 0$, either both y and λ_3 are 0 or both are positive. By (6), both cannot be positive. Therefore, $\lambda_3 = y = 0$.

9. Now $x = 2$, $\lambda_2 = 0$, and $\lambda_1 = \frac{1}{4}$.

10. The final solution is

$$(x, y, \lambda_1, \lambda_2, \lambda_3) = (2, 0, \frac{1}{4}, 0, 0).$$

4 Kuhn-Tucker Formulation

1. The most common constrained maximization problems in economics involve only inequality constraints and a complete set of nonnegativity constraints:

$$\begin{aligned} \max \quad & f(x_1, x_2, \dots, x_n) \\ \text{s.t.} \quad & g_1(x_1, \dots, x_n) \leq b_1, \dots, g_k(x_1, \dots, x_n) \leq b_k \\ & x_1 \geq 0, \dots, x_n \geq 0 \end{aligned}$$

2. If we use the techniques already learned, we would write the Lagrangian as:

$$\begin{aligned} L(\mathbf{x}, \lambda_1, \dots, \lambda_k, \mu_1, \dots, \mu_n) \\ = f(\mathbf{x}) + \lambda_1(b_1 - g_1(\mathbf{x})) + \dots + \lambda_k(b_k - g_k(\mathbf{x})) + \mu_1 x_1 + \dots + \mu_n x_n. \end{aligned}$$

3. And the corresponding first order conditions are:

$$\begin{aligned} \frac{\partial L}{\partial x_1} &= \frac{\partial f}{\partial x_1} - \lambda_1 \frac{\partial g_1}{\partial x_1} - \dots - \lambda_k \frac{\partial g_k}{\partial x_1} + \mu_1 = 0 \\ &\dots\dots\dots \\ \frac{\partial L}{\partial x_n} &= \frac{\partial f}{\partial x_n} - \lambda_1 \frac{\partial g_1}{\partial x_n} - \dots - \lambda_k \frac{\partial g_k}{\partial x_n} + \mu_n = 0 \\ \lambda_1(g_1(\mathbf{x}) - b_1) &= -\lambda_1 \frac{\partial L}{\partial \lambda_1} = 0 \\ &\dots\dots\dots \\ \lambda_k(g_k(\mathbf{x}) - b_k) &= -\lambda_k \frac{\partial L}{\partial \lambda_k} = 0 \\ \mu_1 x_1 &= 0 \\ &\dots\dots\dots \\ \mu_n x_n &= 0 \\ \lambda_1, \dots, \lambda_k, \mu_1, \dots, \mu_n &\geq 0 \\ g_1(x_1, \dots, x_n) &\leq b_1, \dots, g_k(x_1, \dots, x_n) \leq b_k \\ x_1 \geq 0, \dots, x_n &\geq 0. \end{aligned}$$

4. Kuhn and Tucker worked with a Lagrangian $\tilde{L}$ which did not include the nonnegativity constraints:

$$\tilde{L}(\mathbf{x}, \lambda_1, \dots, \lambda_k) = f(\mathbf{x}) + \lambda_1(b_1 - g_1(\mathbf{x})) + \dots + \lambda_k(b_k - g_k(\mathbf{x})).$$

We call $\tilde{L}$ **the Kuhn-Tucker Lagrangian**.

5. Note that

$$L(\mathbf{x}, \lambda_1, \dots, \lambda_k, \mu_1, \dots, \mu_n) = \tilde{L}(\mathbf{x}, \lambda_1, \dots, \lambda_k) + \mu_1 x_1 + \dots + \mu_n x_n.$$

Therefore, for $j = 1, \dots, n$, we have

$$\begin{aligned} \frac{\partial L}{\partial x_j} &= \frac{\partial \tilde{L}}{\partial x_j} + \mu_j = 0 \\ \text{or } \frac{\partial \tilde{L}}{\partial x_j} &= -\mu_j. \end{aligned}$$

6. Since $\mu_j \geq 0$ and $\mu_j x_j = 0$, we obtain

$$\frac{\partial \tilde{L}}{\partial x_j} \leq 0 \text{ and } x_j \frac{\partial \tilde{L}}{\partial x_j} = 0;$$

on the other hand, for any $\mathbf{x}$,

$$\frac{\partial \tilde{L}}{\partial \lambda_j} = \frac{\partial L}{\partial \lambda_j} = b_j - g_j(\mathbf{x}) \geq 0, \text{ and } \lambda_j \frac{\partial \tilde{L}}{\partial \lambda_j} = 0.$$

7. In summary, we find that the first order conditions in terms of the Kuhn-Tucker Lagrangian, or simply, the Kuhn-Tucker conditions, are

$$\begin{aligned} x_j \geq 0, \frac{\partial \tilde{L}}{\partial x_j} \leq 0, \quad x_j \frac{\partial \tilde{L}}{\partial x_j} &= 0, \text{ where } j = 1, \dots, n; \\ \lambda_j \geq 0, \frac{\partial \tilde{L}}{\partial \lambda_j} \geq 0, \quad \lambda_j \frac{\partial \tilde{L}}{\partial \lambda_j} &= 0, \text{ where } j = 1, \dots, k. \end{aligned}$$

8. The Kuhn-Tucker conditions have some advantage over previous first-order conditions. First, it involves $n + k$ equations with $n + k$ unknowns, compared with the $2n + k$ equations in $2n + k$ unknowns. Second, and more important advantage, is the symmetric way that the x and λ enter the first order conditions.

9. Example.

$$\begin{aligned} \max \quad & u = u(x_1, x_2) \\ \text{s.t.} \quad & p_1 x_1 + p_2 x_2 \leq I, \quad x_1, x_2 \geq 0. \end{aligned}$$

10. The Kuhn-Tucker Lagrangian is:

$$\tilde{L}(x_1, x_2, \lambda) = u(x_1, x_2) + \lambda(I - p_1 x_1 - p_2 x_2),$$

and the Kuhn-Tucker conditions are:

$$\begin{aligned} \frac{\partial u}{\partial x_1} - \lambda p_1 &\leq 0, \quad \frac{\partial u}{\partial x_2} - \lambda p_2 \leq 0, \\ x_1 \left(\frac{\partial u}{\partial x_1} - \lambda p_1 \right) &= 0, \quad x_2 \left(\frac{\partial u}{\partial x_2} - \lambda p_2 \right) = 0, \\ \frac{\partial \tilde{L}}{\partial \lambda} &= I - p_1 x_1 - p_2 x_2 \geq 0, \\ \lambda \frac{\partial \tilde{L}}{\partial \lambda} &= \lambda(I - p_1 x_1 - p_2 x_2) = 0, \\ x_1, x_2, \lambda &\geq 0, \end{aligned}$$

5 The Envelope Theorem

After we solve an optimization problem, we can substitute the solution of the problem into the objective function to obtain the maximum-value function. If, in addition, the objective function, constraints and the maximizer are all differentiable with respect to the parameters, then we can apply the *envelope theorem* to analyze the maximum value function.

5.1 Unconstrained Optimization

1. Let $x^*(a)$ be the solution of the maximization problem

$$\max_x f(x, a)$$

where a is a vector of exogenous parameters. (Let's assume that the solution is unique.)

The maximum value function is defined as

$$M(a) \equiv f(x^*(a), a).$$

2. The Envelope Theorem: Assume that f and $x^*(a)$ are differentiable. Then

$$\frac{dM(a)}{da_j} = \frac{\partial f}{\partial a_j}.$$

A change in a_j affect M directly as well as indirectly through $x^*(a)$. The envelope theorem tells us that we can ignore the indirect effect.

3. Proof: Partially differentiating $M(a)$ with respect to a_j , we have

$$\frac{dM(a)}{da_j} = \sum_{i=1}^n \frac{\partial f}{\partial x_i} \frac{\partial x_i}{\partial a_j} + \frac{\partial f}{\partial a_j}.$$

But the first-order condition implies that for all i

$$\frac{\partial f(x^*)}{\partial x_i} = 0.$$

Since f is flat with respect to x at the optimal point x^* , a small change in x^* has no welfare consequence.

The envelope theorem is very useful in economics. It allows us to evaluate the change in M without knowing $\frac{\partial x_i}{\partial a_j}$, the sign of which are difficult to ascertain in many situations.

5.2 Constrained Optimization

1. The same idea applies to the maximum-value function in the constrained optimization.

Let $x^*(a)$ be the solution of the maximization problem

$$\begin{aligned} \max_x f(x, a) \\ \text{s.t. } g(x, a) = 0. \end{aligned}$$

Here, the parameter a may appear in the constraints as well as in the objective function.

2. Theorem: Suppose f , g , and x^* are differentiable with respect to a , then

$$\frac{dM(a)}{da_j} = \frac{\partial \mathcal{L}}{\partial a_j} \Big|_{x(a), \lambda(a)}.$$

It is important to note that here $\frac{dM(a)}{da_j}$ is equal to the partial derivative with respect to $\mathcal{L}$, the Lagrangian function, not f .

3. Example

$$\begin{aligned} \max x_1 x_2 \\ \text{s.t. } I - 2x_1 - 4x_2 = 0. \end{aligned}$$

First forming the Lagrangian:

$$L = x_1x_2 + \lambda(I - 2x_1 - 4x_2)$$

with the first-order conditions:

$$\begin{aligned} L_1 &= x_2 - 2\lambda = 0 \\ L_2 &= x_1 - 4\lambda = 0 \\ L_\lambda &= I - 2x_1 - 4x_2 = 0 \end{aligned}$$

Solving them we get:

$$x_1 = \frac{I}{4}, \quad x_2 = \frac{I}{8}, \quad \lambda = \frac{I}{16}.$$

We then form the maximum value function

$$M(I) = x_1(I)x_2(I) = \frac{I^2}{32}$$

Therefore we have

$$\frac{dM(I)}{dI} = \frac{\partial L}{\partial I} = \lambda = \frac{I}{16}$$

6 Second Order Conditions

We know that the second order condition for maximizing an unconstrained function $f(\mathbf{x})$ is that $H(\mathbf{x}^*)$ is negative definite.

6.1 Equality constraints

1. For the problem with two variables and one constraint such that

$$\max_{x_1, x_2} f(x_1, x_2) \quad \text{subject to } g(x_1, x_2) = 0,$$

2. The sufficient condition is: if $(x_1^*, x_2^*, \lambda^*)$ solves the first-order conditions, and if $|\overline{H}| > 0 (< 0)$ when evaluated at $(x_1^*, x_2^*, \lambda^*)$, then (x_1^*, x_2^*) is a local maximum (minimum), where

$$\overline{H} = \begin{pmatrix} 0 & g_1 & g_2 \\ g_1 & L_{11} & L_{12} \\ g_2 & L_{21} & L_{22} \end{pmatrix}.$$

3. For the problem with n variables and $m < n$ constraints such as

$$\begin{aligned} &\max \text{ or } \min f(x_1, \dots, x_n) \\ &s.t. \quad g^1(\mathbf{x}) = 0, \dots, g^m(\mathbf{x}) = 0, \end{aligned}$$

let $(\mathbf{x}^*, \boldsymbol{\lambda}^*)$ solve the first-order conditions.

4. (J&R, Theorem A2.18). $\mathbf{x}^*$ is the local maximum if the last $n - m$ leading principal minors alternate in sign, beginning with that of $(-1)^{m+1}$ when evaluate at $(\mathbf{x}^*, \boldsymbol{\lambda}^*)$, where

$$\overline{H}_k = \begin{pmatrix} 0 & \dots & 0 & g_1^1 & \dots & g_k^1 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & \dots & 0 & g_1^m & \dots & g_k^m \\ g_1^1 & \dots & g_1^m & L_{11} & \dots & L_{1k} \\ \dots & \dots & \dots & \dots & \dots & \dots \\ g_k^1 & \dots & g_k^m & L_{k1} & \dots & L_{kk} \end{pmatrix}, \text{ where } k = m + 1, \dots, n.$$

5. $\mathbf{x}^*$ is the local minimum if the last $n - m$ leading principal minors are all have the same sign as $(-1)^m$, when evaluate at $(\mathbf{x}^*, \boldsymbol{\lambda}^*)$.

6.2 Inequality constraints

- Given a solution $(\mathbf{x}^*, \boldsymbol{\lambda}^*)$ of the first-order conditions, we divide the inequality constraints into binding constraints and nonbinding constraints at $\mathbf{x}^*$. On the one hand, we treat the binding constraints like equality constraints; on the other hand, the multipliers for the nonbinding constraints must be 0 and these constraints drop out of the Lagrangian.
- The problem is as follows:

$$\begin{aligned} & \max f(\mathbf{x}) \\ & s.t. \ g_1(\mathbf{x}) \leq b_1, \dots, g_m(\mathbf{x}) \leq b_m \\ & \quad h_1(\mathbf{x}) = c_1, \dots, h_k(\mathbf{x}) = c_k. \end{aligned}$$

- Suppose $(\mathbf{x}^*, \boldsymbol{\lambda}^*)$ solve the first-order conditions. For notation purpose, suppose that $g_1, \dots, g_e$ are binding at $\mathbf{x}^*$ and $g_{e+1}, \dots, g_m$ are not binding.
- $\mathbf{x}^*$ is the local maximum if the last $n - (e + k)$ leading principal minors alternate in sign, beginning with that of $(-1)^{e+k+1}$, when evaluate at $(\mathbf{x}^*, \boldsymbol{\lambda}^*)$, where

$$\overline{H}_i = \begin{pmatrix} 0 & \dots & 0 & 0 & \dots & 0 & \frac{\partial g_1}{\partial x_1} & \dots & \frac{\partial g_1}{\partial x_i} \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & \dots & 0 & 0 & \dots & 0 & \frac{\partial g_e}{\partial x_1} & \dots & \frac{\partial g_e}{\partial x_i} \\ 0 & \dots & 0 & 0 & \dots & 0 & \frac{\partial h_1}{\partial x_1} & \dots & \frac{\partial h_1}{\partial x_i} \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & \dots & 0 & 0 & \dots & 0 & \frac{\partial h_k}{\partial x_1} & \dots & \frac{\partial h_k}{\partial x_i} \\ \frac{\partial g_1}{\partial x_1} & \dots & \frac{\partial g_e}{\partial x_1} & \frac{\partial h_1}{\partial x_1} & \dots & \frac{\partial h_k}{\partial x_1} & \frac{\partial^2 L}{\partial x_1^2} & \dots & \frac{\partial^2 L}{\partial x_1 \partial x_i} \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ \frac{\partial g_1}{\partial x_i} & \dots & \frac{\partial g_e}{\partial x_i} & \frac{\partial h_1}{\partial x_i} & \dots & \frac{\partial h_k}{\partial x_i} & \frac{\partial^2 L}{\partial x_1 \partial x_i} & \dots & \frac{\partial^2 L}{\partial x_i^2} \end{pmatrix}, \text{ where } i = (e+k)+1, \dots, n.$$

5. The minimization problem is as follows:

$$\begin{aligned} \min \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & g_1(\mathbf{x}) \geq b_1, \dots, g_m(\mathbf{x}) \geq b_m \\ & h_1(\mathbf{x}) = c_1, \dots, h_k(\mathbf{x}) = c_k. \end{aligned}$$

6. Suppose $(\mathbf{x}^*, \boldsymbol{\lambda}^*)$ solve the first-order conditions. Suppose also that $g_1, \dots, g_e$ are binding at $\mathbf{x}^*$ and $g_{e+1}, \dots, g_m$ are not binding.
7. $\mathbf{x}^*$ is the local minimum if the last $n - (e + k)$ leading principal minors are all have the same sign as $(-1)^{e+k}$ when evaluate at $(\mathbf{x}^*, \boldsymbol{\lambda}^*)$.

7 References

1. JR, Chapter A2
2. MWG: Mathematical Appendix